

GitStack Connect - Quick Reference Guide

Project Overview

GitStack Connect is a developer intelligence platform that analyzes GitHub and StackOverflow profiles to build knowledge graphs, detect communities, and provide personalized repository recommendations.

Key Technologies

Frontend

- React 19.2.0
- Cytoscape.js (graph visualization)
- Axios (HTTP client)

Backend

- FastAPI (REST API)
 - NetworkX (graph algorithms)
 - Node2Vec (graph embeddings)
 - Sentence Transformers (NLP)
 - Groq LLM (AI features)
 - SQLite (database)
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Core Algorithms

1. Node2Vec Embeddings

- **Purpose:** Convert graph to vector representations
- **Dimensions:** 64
- **Use:** Similarity computation for recommendations

2. Louvain Community Detection

- **Purpose:** Identify developer communities
- **Method:** Modularity optimization
- **Output:** Community IDs for each node

3. Centrality Metrics

- **Degree:** Number of connections
- **Betweenness:** Bridge role in network
- **Closeness:** Average distance to others
- **Influence:** Composite score (0-100)

4. Recommendation Pipeline

1. Profile-based candidate filtering
2. Node2Vec cosine similarity
3. Jaccard coefficient (topology)
4. Profile boost (language/topic match)
5. LLM-based final ranking

5. Project Ideas Search

- **Sources:** GitHub repositories, StackOverflow questions, Dev.to articles
- **Method:** Parallel API calls with independent pagination
- **Query:** User enters any project idea, technology, or topic
- **Results:** Unified display with separate sections for each source

API Endpoints

Endpoint	Method	Description
/enrich/{username}	GET	Enrich GitHub profile
/recommend/{node_id}	GET	Get recommendations
/predict/{user_id}	GET	Predict role
/metrics/{node_id}	GET	Network metrics
/search/project	GET	Search project ideas across GitHub, StackOverflow, and Dev.to

Data Flow

User Input → GitHub API → Data Processing → Graph Building → Storage

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Recommendations ← LLM Ranking ← Scoring ← Embeddings ← Graph

Project Search: Query → Parallel API Calls (GitHub, SO, Dev.to) → Unified Results

Key Features

1. **Profile Enrichment:** GitHub + StackOverflow data aggregation
2. **Network Visualization:** Interactive graph with Cytoscape.js
3. **Community Detection:** Louvain algorithm with visual highlighting
4. **Recommendations:** Multi-stage personalized repository suggestions

5. **Role Prediction:** AI + rule-based classification
 6. **Network Metrics:** Centrality measures with explanations
 7. **Project Ideas Search:** Multi-source search (GitHub repos, StackOverflow questions, Dev.to articles)
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File Structure

```

backend/
  app.py           # FastAPI application
  enrich.py        # Profile enrichment
  recommendation.py # Recommendation engine
  graph_ops.py     # Graph algorithms
  nlp_ops.py       # NLP operations
  llm.py           # LLM integration
  storage.py       # Database operations
  fetchers/        # API clients

frontend/src/
  App.js           # Main application
  components/
    ProfileCard.jsx
    GraphView.jsx
    TopRecommendations.jsx
    ProjectResults.jsx
  api.js           # API client

```

Important Formulas

Influence Score

$$\text{Influence} = (\text{Degree} \times 0.4 + \text{Betweenness} \times 0.4 + \text{Closeness} \times 0.2) \times 100$$

Recommendation Score

$$\text{Score} = \alpha \times \text{Cosine_Similarity} + (1 - \alpha) \times \text{Jaccard} + \text{Profile_Boost}$$

where $\alpha = 0.7$

Jaccard Coefficient

$$J(A,B) = |A \cap B| / |A \cup B|$$

Configuration

Environment Variables

- **GITHUB_API_TOKEN:** GitHub API token (optional)
- **GROQ_API_KEY:** Groq API key (required for AI features)
- **GEMINI_API_KEY:** Gemini API key (optional, fallback)

Default Ports

- **Backend:** `http://localhost:8000`
 - **Frontend:** `http://localhost:3000`
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Performance Metrics

- **Enrichment:** 5-15 seconds per user
 - **Recommendations:** 1-3 seconds
 - **Metrics:** <1 second
 - **Project Search:** 1-2 seconds (parallel API calls)
 - **Graph Rendering:** Real-time
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Common Use Cases

1. **Discover Projects:** Find repositories matching your skills
 2. **Network Analysis:** Understand your position in developer community
 3. **Community Detection:** Identify developer clusters
 4. **Profile Analysis:** Get comprehensive developer insights
 5. **Role Prediction:** Automatically classify developer roles
 6. **Project Ideas Search:** Enter any query to find projects, solutions, and articles from GitHub, StackOverflow, and Dev.to
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Troubleshooting

Issue: API Quota Exceeded

- **Solution:** System automatically falls back to rule-based methods

Issue: No Recommendations

- **Check:** User must have enriched profile in database
- **Check:** Graph must have sufficient nodes

Issue: Slow Performance

- **Solution:** Embeddings are cached, first run may be slower
 - **Solution:** Limit candidate pool size
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Future Enhancements

- More data sources (GitLab, Bitbucket)
- Graph database migration (Neo4j)
- Real-time updates
- Mobile application
- Advanced analytics dashboard